Sravya Aravapalli

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Education

Carnegie Mellon University **May 2028**
Bachelor of Science in Computer Science, Minor in Machine Learning *Pittsburgh, PA*
Courses: *15-213 Intro to Computer Systems, 15-210 Parallel Data Structures & Algorithms, 15-122 Imperative Computation, 15-150 Functional Programming, 15-151/251 Discrete Mathematics, 10-301 Intro to ML, 15-259 Probability & Computing, 21-241 Linear Algebra, 21-266 Vector Calculus*
Organizations: *Carnegie Mellon Racing, Society of Women Engineers, Alpha Chi Omega: Kappa Nu*

Technical skills

Languages: Python, Java, C++, C, TypeScript, JavaScript, SML, HTML/CSS
Technologies: Git, NumPy, PyTorch, Pandas, Sklearn, React, Next.js, PostgreSQL, Prisma, Tailwind CSS

Experience

Axon **May 2026 – Present**
Firmware Engineering Intern *Boston, MA*

- Developing C/C++ firmware as part of the Platform Firmware team to enable LTE-based offloading and transmission of embedded device sensor data, enhancing data accessibility
- Building Python automation scripts to analyze IMU data streams and extract key metrics

Carnegie Mellon University **Aug. 2026 (Incoming)**
Teaching Assistant *Pittsburgh, PA*

- Will lead recitations and office hours, assist in developing and grading assignments and exams, and support review sessions for 15-151 Mathematical Foundations for Computer Science, a proof-based discrete mathematics course

Research experience

CMU Computer Science Department **Jan. 2026 – May 2026**
Undergraduate Researcher *Pittsburgh, PA*

- Led the migration and redesign of Gamebot's C++ Interaction Manager to Python for an autonomous Scrabble-playing robot used in HCI research under Professor Reid Simmons

University of Minnesota **June 2025 – Aug. 2025**
Undergraduate Researcher *Minneapolis, MN*

- Selected into the 'AI for Earth Summer School' and developed LSTM-based models to predict carbon flux metrics including NEE, GPP, and Reco across global environmental datasets
- Achieved an RMSE of 0.53 for GPP prediction, corresponding to around 5% of its daily flux range

SBPL Research Lab, CMU Robotics Institute **Feb. 2025 – Apr. 2025**
Undergraduate Researcher *Pittsburgh, PA*

- Built and optimized a 2D occupancy grid–based collision checker using rectangular footprints for quadrupedal motion planning research under Rishi Veerapaneni and Professor Maxim Likhachev
- Implemented batched geometric transformations and collision validation using NumPy vectorization (einsum, broadcasting) and linear interpolation for efficient path checks

Projects

SustainaTree (TartanHacks26) | *React Native, Expo, Firebase, TypeScript* **Feb. 2026**

- Built a real-time sustainability tracking app with dynamic tree-growth visualization driven by user eco-actions, AI-assisted event autofill, and Firebase-based user data synchronization

Astro Sandbox | *TypeScript, Next.js, React, Three.js, PostgreSQL* **Aug. 2025 – Nov. 2025**

- Developed a real-time orbital mechanics simulator with interactive 3D Three.js visualization, Web Worker–based physics computation, and Prisma ORM/PostgreSQL backend APIs